PCCST501 – COMPUTER NETWORKS

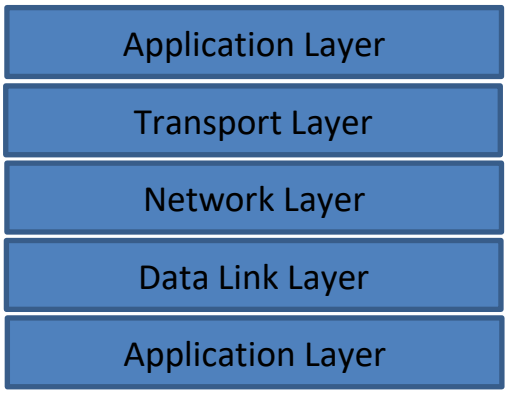
MODULE 1 ASSIGNMENT

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How the Internet Delivers a Web Page

Website Choosen: Instagram(www.instagram.com)

1. TCP/IP Protocol Stack and Role of Each Layer



Data travels down the stack on the sender's side and up the stack on the receiver's side.

Application Layer: The Application Layer provides network services directly to user applications such as Instagram. It uses protocols like HTTPS, HTTP, and DNS to enable communication between the user's browser and Instagram's servers. This layer is responsible for sending requests to Instagram's servers and receiving webpages, images, videos, and other content.

Transport Layer: The Transport Layer uses the TCP (Transmission Control Protocol) to provide reliable communication between the browser and Instagram's servers. It divides the data into smaller segments before transmission and reassembles them at the destination. It also ensures that all data reaches the browser correctly, in the proper order, and without errors.

Network Layer: The Network Layer uses IP (Internet Protocol) addresses to identify the source and destination devices. It determines the best path for transmitting packets across different networks and routes them from the user's device to Instagram's servers.

Data Link Layer: The Data Link Layer is responsible for transferring data in the form of frames over the local network using Wi-Fi or Ethernet. It uses MAC addresses to identify devices on the same network and ensures reliable communication between directly connected devices.

Physical Layer: The Physical Layer converts digital data into electrical, optical, or wireless signals for transmission. It sends these signals through physical media such as network cables, optical fibers, or Wi-Fi, enabling the actual transfer of data between devices.

2. Role of DNS (Domain Name System)

DNS converts the website name into an IP address.

www.instagram.com --> DNS Server --> 157.240.xxx.xxx (Example IP Address)

Process:

1. The user types www.instagram.com into the browser.
2. The browser checks its DNS cache.
3. If the address is not available, it sends a DNS query.
4. The DNS server returns Instagram's IP address.
5. The browser connects to Instagram's web server using the IP address.

3. How HTTP Requests and Receives a Webpage

HTTP is used for communication between the browser and web server.

Steps:

1. Browser sends an HTTP GET request.
2. Server receives the request.
3. Server processes it.
4. Server sends an HTTP Response containing:
    - HTML
    - Images
    - CSS
    - JavaScript
5. Browser displays the webpage.

4. How TCP Provides Reliable Communication

HTTP uses TCP because webpages must be delivered correctly.

TCP provides:

- Connection establishment (Three-Way Handshake)

  Client Server

  SYN -------->

  <------ SYN + ACK

  ACK -------->

  After connection establishment, HTTP data is transferred safely.

- Error detection
- Packet sequencing

- Retransmission of lost packets
- Flow control
- Congestion control

5. Client–Server or Peer-to-Peer?

Instagram follows the Client–Server Architecture.

Client

- User's web browser or Instagram mobile app.

Server

- Instagram's web servers and data centers.

Justification

- The client sends requests to Instagram.
- Instagram's servers process login requests, fetch posts, reels and messages.
- The server sends the requested information back to the client.
- Communication is centralized.

Therefore, Instagram uses the Client–Server model.

6. Difference Between HTTP and FTP

HTTP	FTP
Used for web pages	Used for file transfer
Uses port 80 (HTTP)	Uses port 21
Displays web content	Uploads and downloads files
Stateless protocol	Supports authenticated file management
Mainly browser-based	Uses FTP client software

7. Cookies and User Experience

Cookies are small files stored in the user's browser.

They improve user experience by remembering:

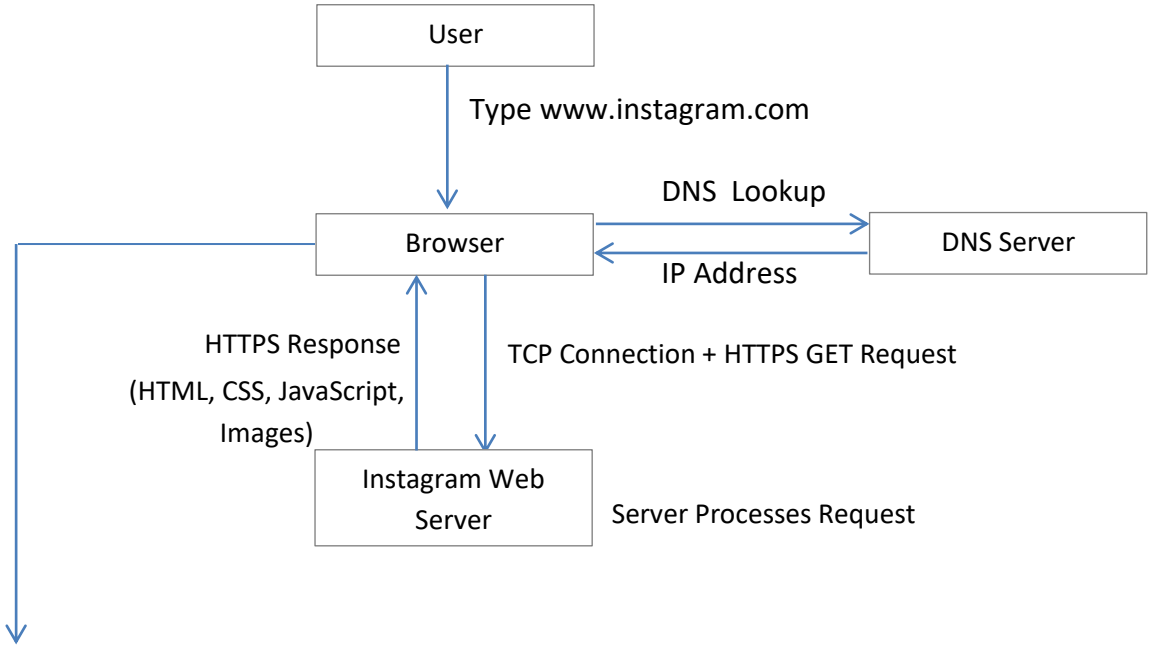
- Login information
- User preferences

- Shopping cart contents
- Language settings

Real-world Example

When you log in to Instagram and select "Remember Me", a cookie stores your session information. The next time you visit Instagram, you remain logged in without entering your username and password again. This provides a faster, more secure and personalized browsing experience.

8. Flow Diagram



Displays Instagram Page

9. Conclusion

Application-layer protocols play an essential role in everyday Internet communication. When a user accesses a website, protocols such as DNS, HTTP, and TCP work together to ensure that the request reaches the correct server and that the webpage is delivered accurately and efficiently. DNS translates easy-to-remember domain names into IP addresses, HTTP enables communication between browsers and web servers, and TCP guarantees reliable and error-free data transmission. Cookies further improve the browsing experience by remembering user preferences and login sessions. Without these application-layer protocols, accessing websites, sending emails, streaming videos, or using online services would not be possible. They form the foundation of modern Internet communication and make web applications reliable, fast, and user-friendly.